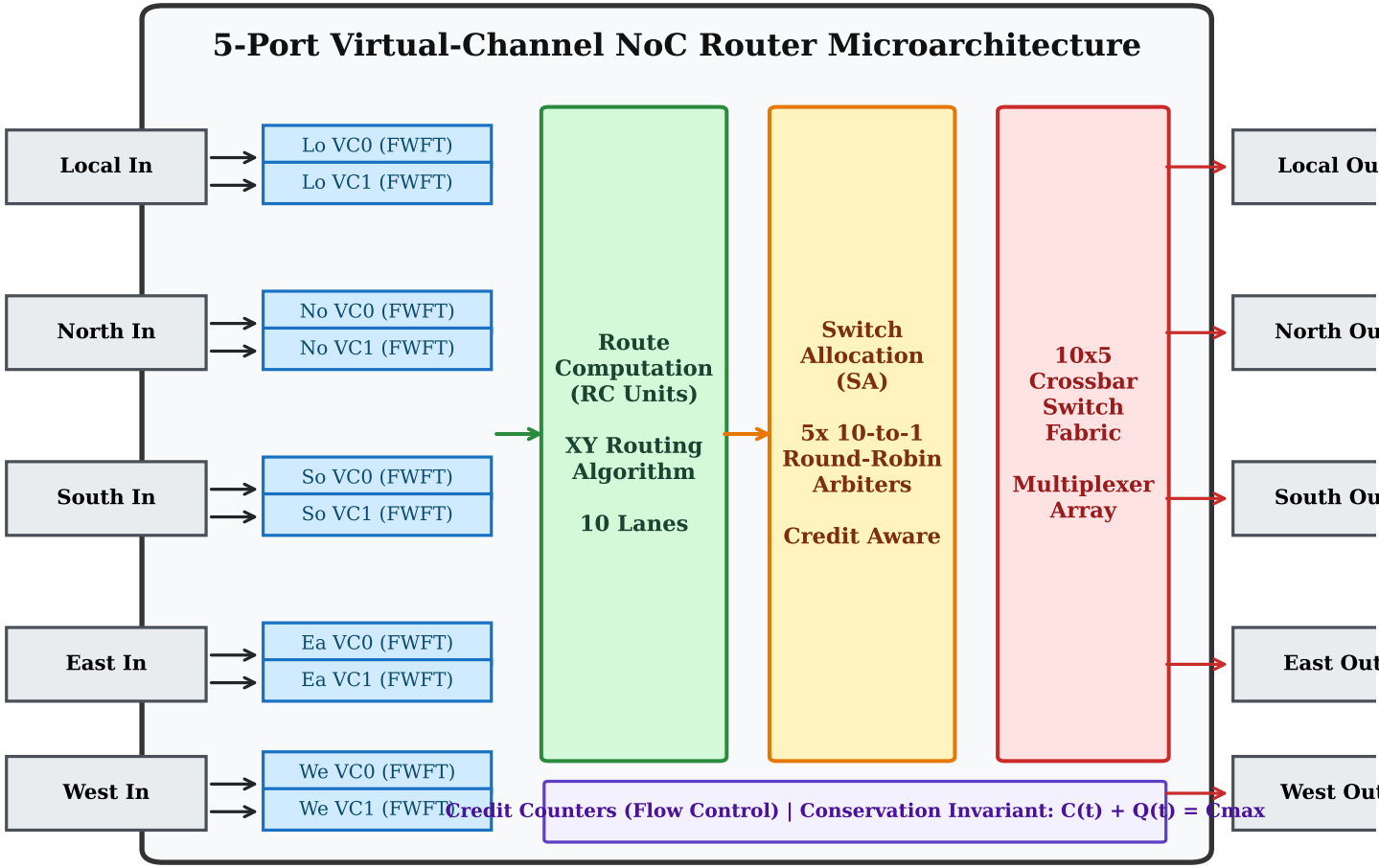
5-Port Virtual-Channel NoC Router Microarchitecture
Local In
North In
South In
East In
West In
Lo VC0 (FWFT)
Lo VC1 (FWFT)
No VC0 (FWFT)
No VC1 (FWFT)
So VC0 (FWFT)
So VC1 (FWFT)
Ea VC0 (FWFT)
Ea VC1 (FWFT)
We VC0 (FWFT)
We VC1 (FWFT)
Route Computation (RC Units)
XY Routing Algorithm
10 Lanes
Switch Allocation (SA)
5x 10-to-1 Round-Robin Arbiters
Credit Aware
10x5 Crossbar Switch Fabric
Multiplexer Array
Local Out
North Out
South Out
East Out
West Out
Credit Counters (Flow Control) | Conservation Invariant: C(t) + Q(t) = Cmax